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Roll No

CE-702(A)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Prestressed Concrete Structure

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. A rectangular concrete beam, 100 mm wide by 250 mm deep, spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm. The beam supports a live load of 1.2 kN/m.
 - a) Calculate the resultant stress distribution for the central cross section of the beam. The density of concrete is 24 kN/m³.
 - b) Find the magnitude of the prestressing force with an eccentricity of 40 mm which can balance the stresses due to dead and live loads at the bottom fibre of the central section of the beam.
2. a) A pre-tensioned prestressed concrete beam having a rectangular section, 300 mm wide and 500mm deep has an effective cover of 40 mm. If $f_{ck} = 40 \text{ N/mm}^2$, $f_p = 1600 \text{ N/mm}^2$ and the area of prestressing steel $A_p = 461 \text{ mm}^2$. Calculate the ultimate flexural strength of the section using IS:1343 code provisions.

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- b) Why is the high strength of concrete and high grade of steel required for prestressed concrete? Explain in detail.
- 3. a) A rectangular beam 250×500 mm in section is simply supported over a span of 10m. It is prestressed with a parabolic cable which has a maximum eccentricity of 200 mm at midspan and 40 mm at support sections. Effective prestressing force is 1450 kN. Concrete grade is M40. Determine the deflection due to prestress and self-weight.
b) What is the anchorage zone?
- 4. a) What is relaxation of stress in steel? How do you account for it in prestressed members? Explain the provisions made in IS: 1343 for relaxation loss.
b) Explain the various methods of achieving continuity of PSC members.
- 5. a) Explain Load Balancing Concept with suitable examples.
b) Explain the objectives and methods of partial prestressing.
- 6. A prestressed concrete beam $400 \text{ mm} \times 400 \text{ mm}$, is prestressed by 10 wires, each of 8 mm diameter. The wires are initially stressed to 1200 N/mm^2 with their centroids located 60 mm from the soffit. Calculate the final percentage loss of stress due to elastic deformation, creep, shrinkage and relaxation using the following data: $E_s = 210 \text{ kN/mm}^2$ and $E_c = 32 \text{ kN/mm}^2$, Creep coefficient = 1.6, Residual shear strain = 3×10^{-4} Relaxation of steel stress = 90 N/mm^2 .

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7. A continuous PSC beam PQR ($PQ = QR = 10\text{ m}$) has a uniform rectangular section with a width of 100 mm and depth of 300 mm. The cable carrying an effective prestressing force of 360 kN is parallel to the axis of the beam and located at 300 mm from the soffit. Determine the secondary and resultant moment at the central support. If the beam supports an imposed load of 1.5 kN/m, calculate the resultant stresses at top and bottom of the beam at B.
8. Explain the following:
- i) Difference between pre-tensioned and post-tensioned members.
 - ii) Advantages of composite construction.
 - iii) Explain any two methods of prestressing systems.
 - iv) Factors influencing deflections of PSC members.
